Experiment 2 : Interacting with Web APIs

Aim: To interact with a weather API to retrieve weather data for a specific location and perform d modeling and visualization to analyze weather patterns over time.

In [1]:
```
# A52 Dipanshu Ambilkar
import requests
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

In [2]:
```
url = "https://api.open-meteo.com/v1/forecast"
p = {"latitude": 18.52, "longitude": 73.85, "hourly": "temperature_2m,rel
r = requests.get(url, params=p, timeout=30).json()
print("API response keys:", list(r.keys()))
```

```
API response keys: ['latitude', 'longitude', 'generationtime_ms',
'utc_offset_seconds', 'timezone', 'timezone_abbreviation', 'elevation',
'hourly_units', 'hourly']
```

In [3]:
```
h = r["hourly"]
df = pd.DataFrame({
    "time": pd.to_datetime(h["time"]),
    "temp": h["temperature_2m"],
    "humidity": h["relative_humidity_2m"],
    "wind": h["wind_speed_10m"],
    "rain": h["precipitation"],
})
df.head()
```

Out[3]:

	time	temp	humidity	wind	rain
0	2026-11-01 00:00:00	22.5	80	10.2	0.0
1	2026-11-01 01:00:00	22.1	83	7.1	0.0
2	2026-11-01 02:00:00	22.8	82	7.1	0.0
3	2026-11-01 03:00:00	24.3	76	8.3	0.0
4	2026-11-01 04:00:00	26.2	69	11.2	0.0

In [4]:
```
print("shape:", df.shape)
print("missing values:")
print(df.isnull().sum())
```

```
shape: (168, 5)
missing values:
time        0
temp        0
humidity    0
wind        0
rain        0
dtype: int64
```

In [5]:
```
df = df.dropna()
print("rows after dropna:", len(df))
```

```
rows after dropna: 168
```

In [6]:
```
print("Avg temp:", df["temp"].mean())
```

```
Avg temp: 24.383333333333333
```

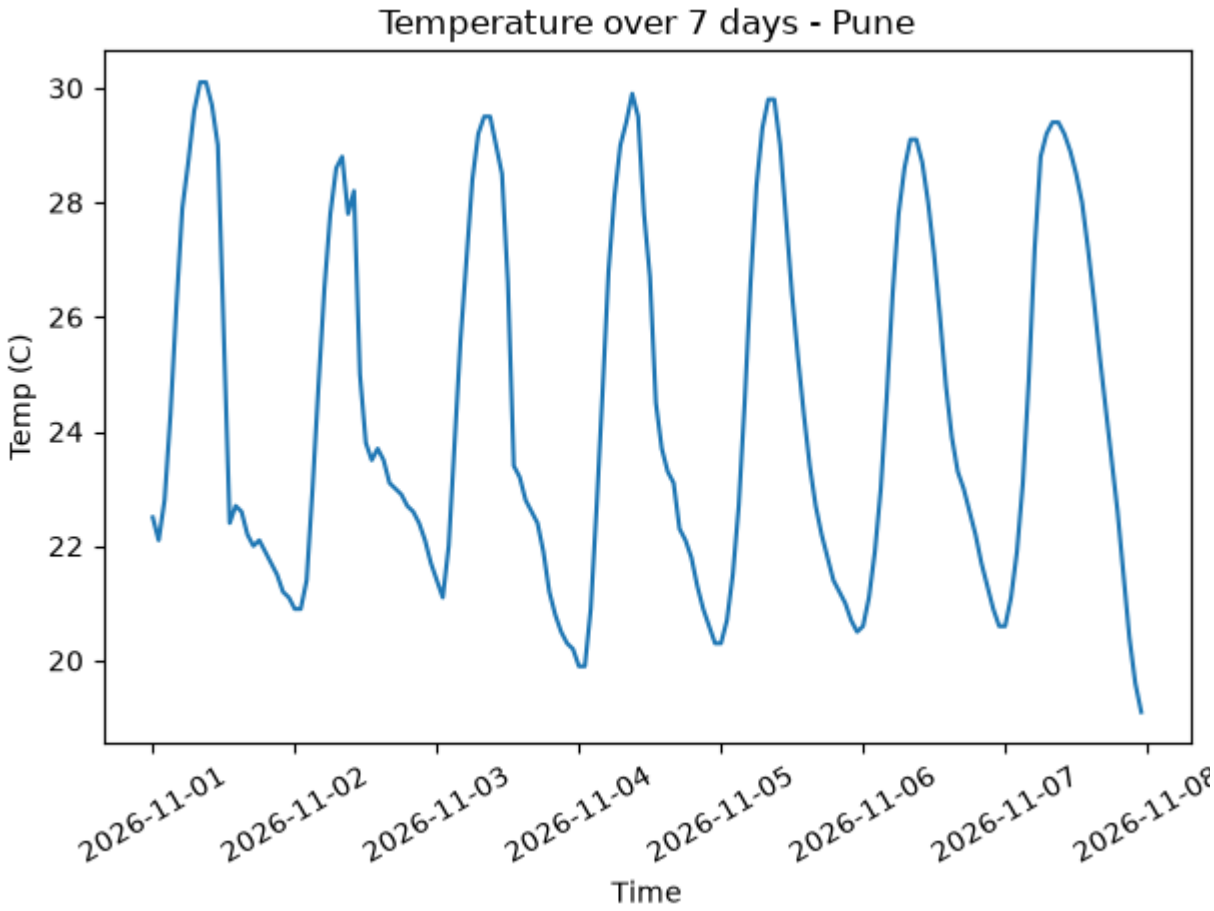
In [7]:
```
print("Max temp:", df["temp"].max(), " Min temp:", df["temp"].min())
```

```
Max temp: 30.1  Min temp: 19.1
```

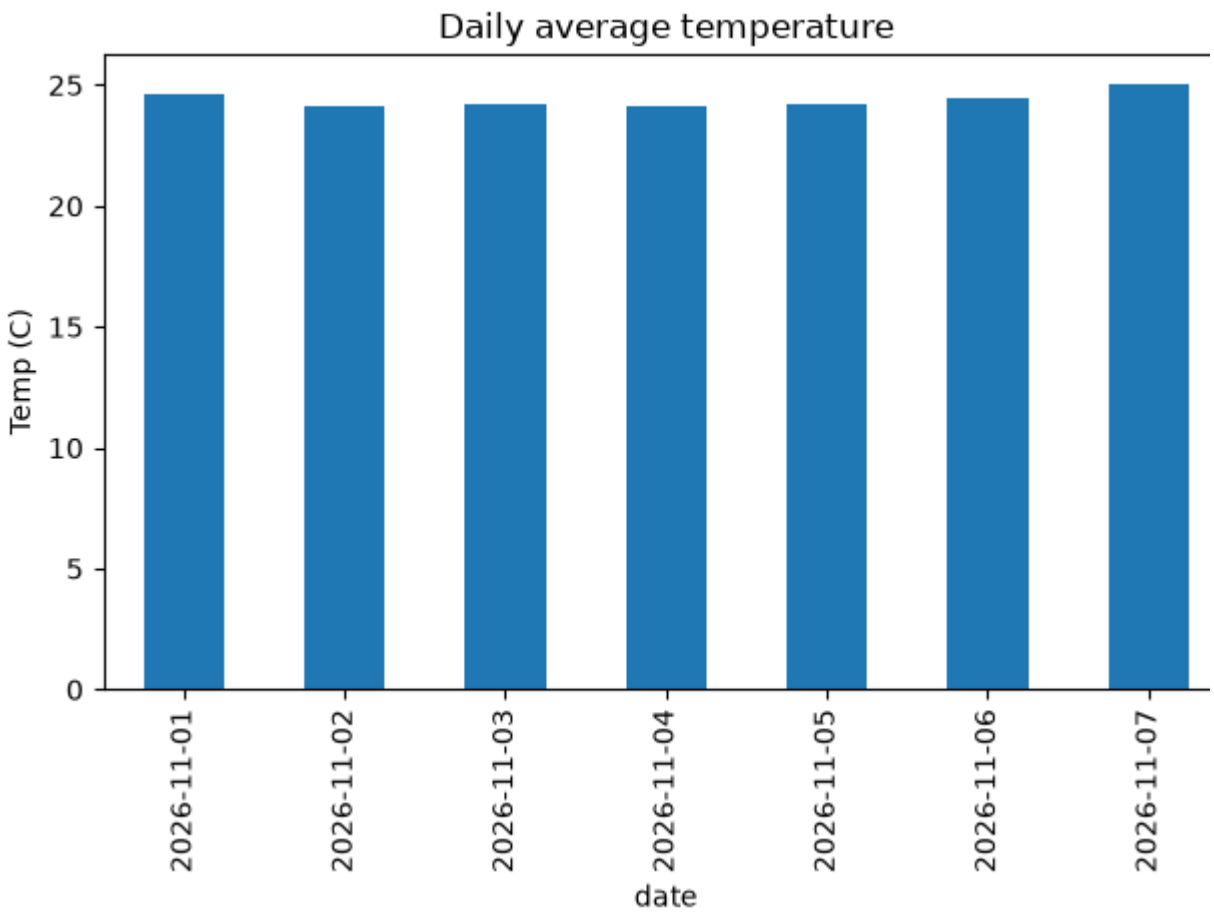
In [8]:
```
df["date"] = df["time"].dt.date
daily = df.groupby("date").agg({"temp": "mean", "humidity": "mean", "wind
print(daily)
```

```
                temp   humidity      wind  rain
date
2026-11-01  24.587500  78.041667  6.300000   7.4
2026-11-02  24.112500  84.000000  5.125000   3.1
2026-11-03  24.204167  80.750000  5.858333   0.9
2026-11-04  24.133333  80.833333  5.641667   1.0
2026-11-05  24.212500  78.916667  7.683333   0.3
2026-11-06  24.429167  77.916667  9.833333   1.2
2026-11-07  25.004167  66.583333  7.312500   0.0
```

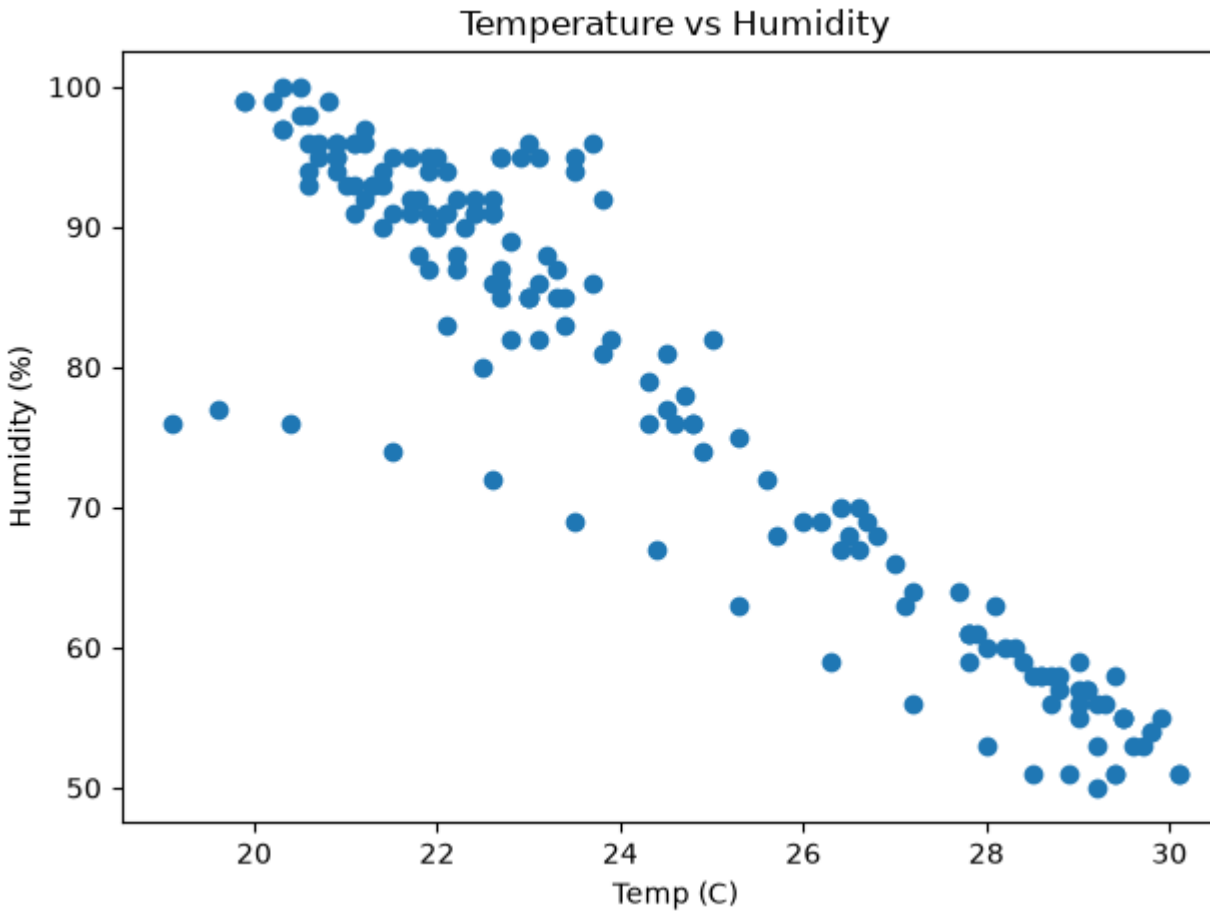
In [9]:
```
plt.plot(df["time"], df["temp"])
plt.title("Temperature over 7 days - Pune")
plt.xlabel("Time")
plt.ylabel("Temp (C)")
plt.xticks(rotation=30)
plt.tight_layout()
plt.show()
```



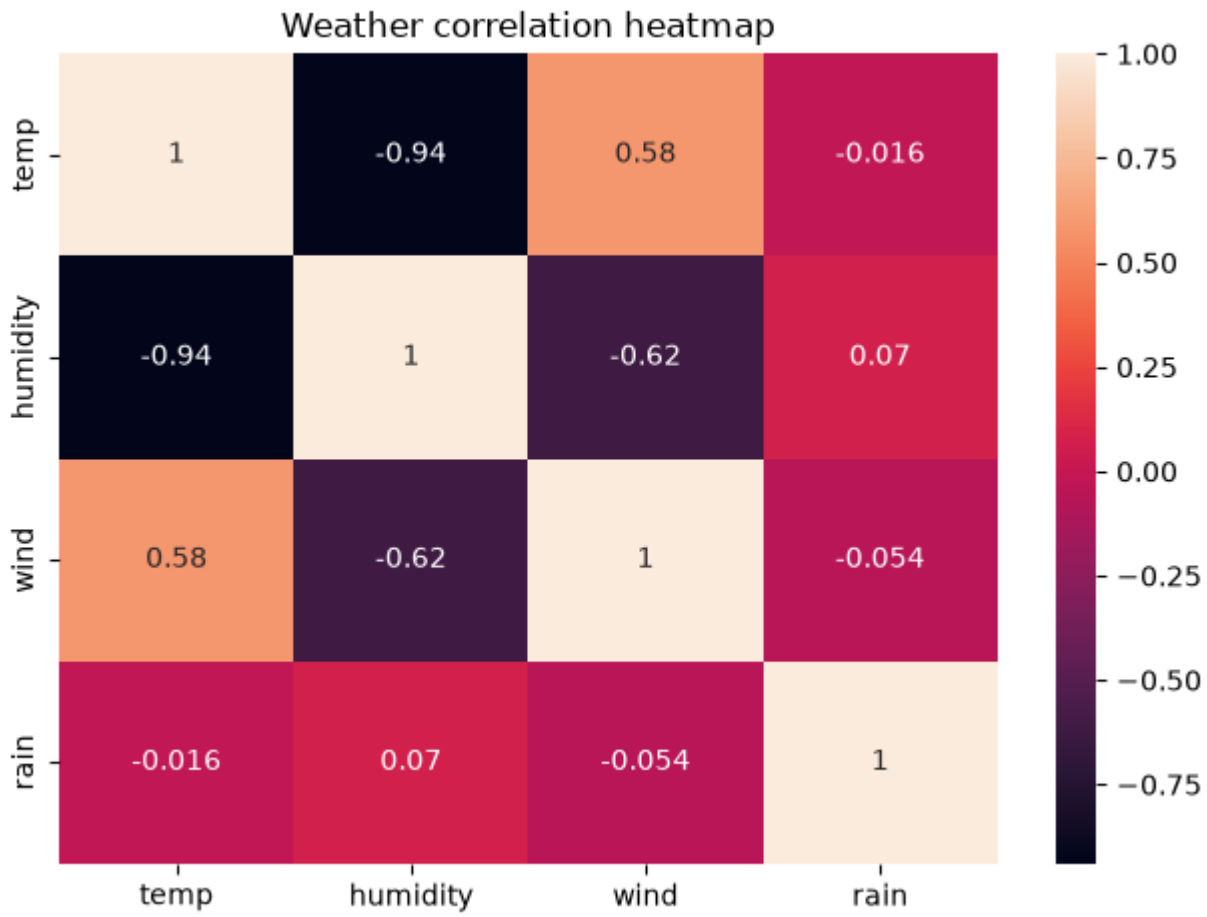
In [10]:
```
daily["temp"].plot(kind="bar")
plt.title("Daily average temperature")
plt.ylabel("Temp (C)")
plt.tight_layout()
plt.show()
```



In [11]:
```
plt.scatter(df["temp"], df["humidity"])
plt.title("Temperature vs Humidity")
plt.xlabel("Temp (C)")
plt.ylabel("Humidity (%)")
plt.tight_layout()
plt.show()
```



In [12]:
```
sns.heatmap(df[["temp", "humidity", "wind", "rain"]].corr(), annot=True)
plt.title("Weather correlation heatmap")
plt.tight_layout()
plt.show()
```



In [13]:
```
print("done")
```

```
done
```